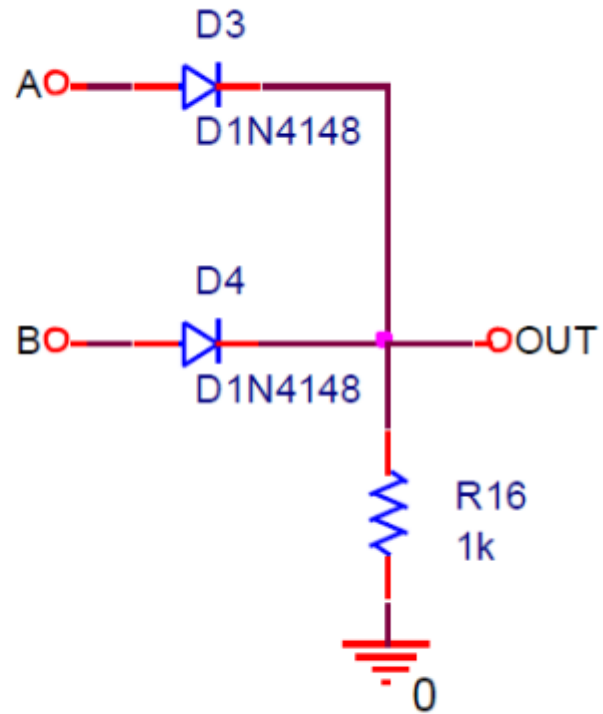


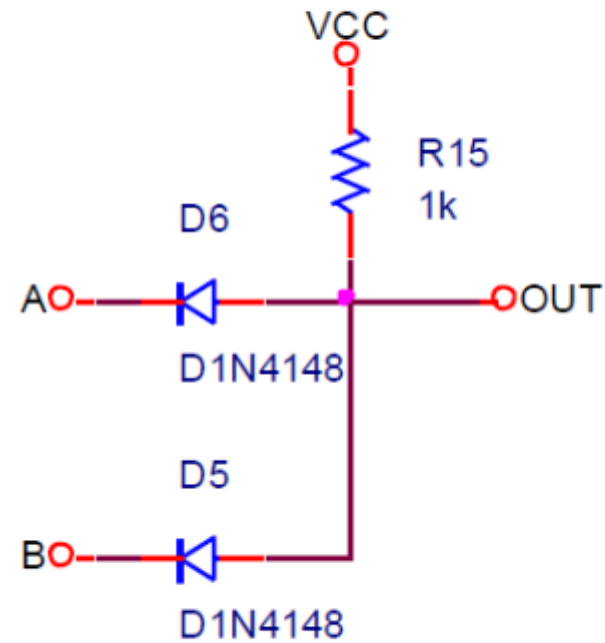
# HW11

엄민호

# 1. Two-Circuit Comparison ( $V_{CC}=5V$ , $A \cdot B = H/L$ )

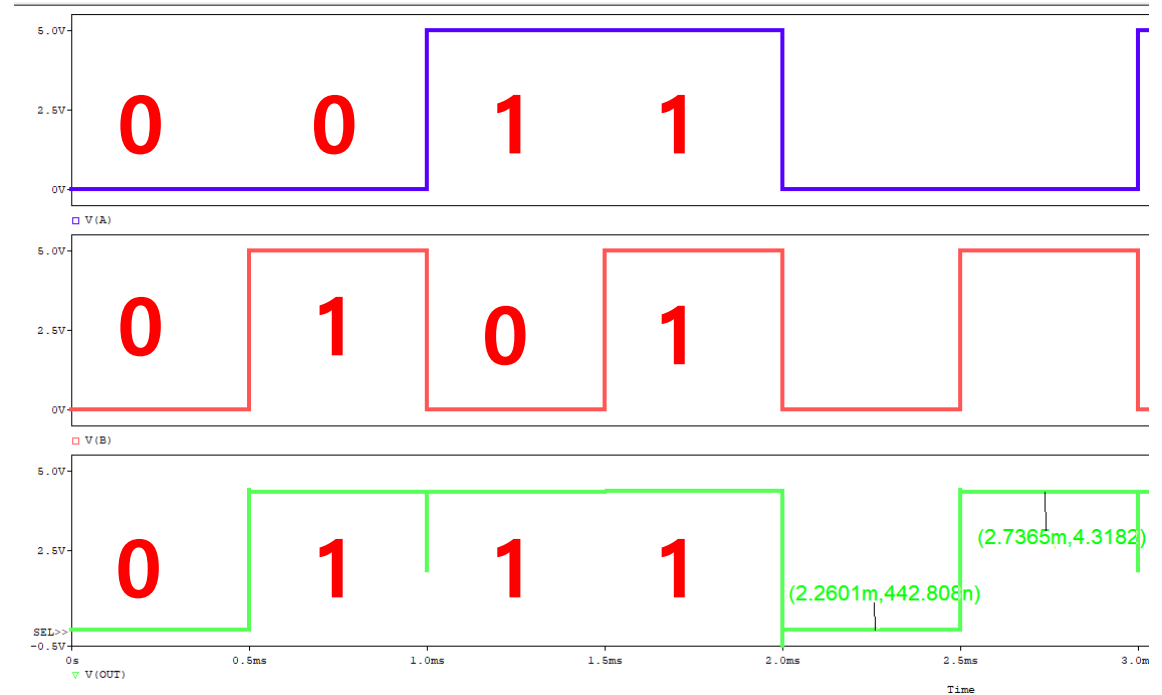
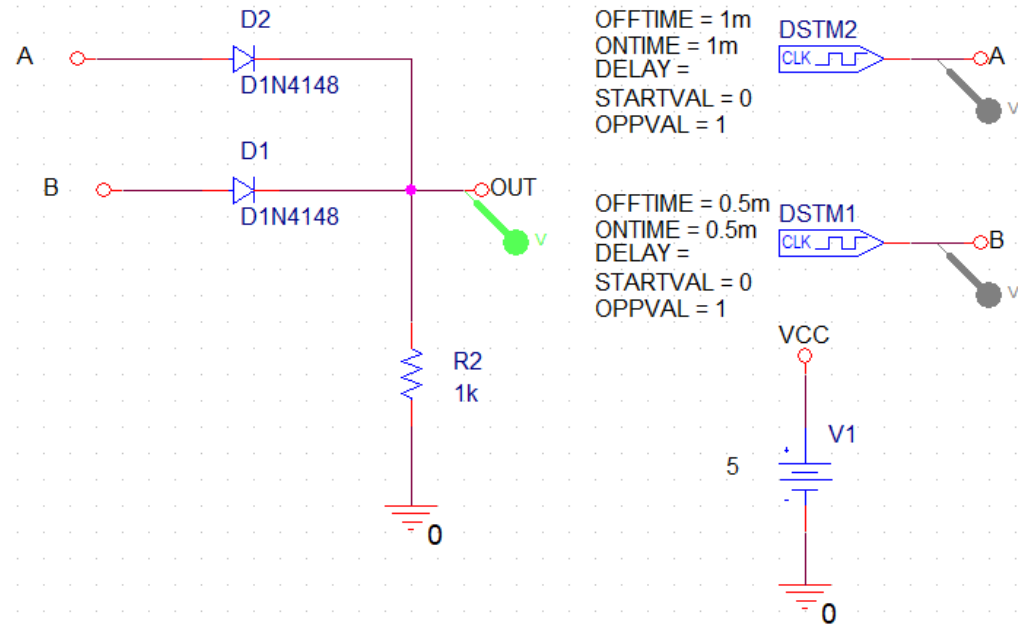


<OR-ing>



<AND-ing>

# 1-(1) OR GATE Design with diodes and Simulation

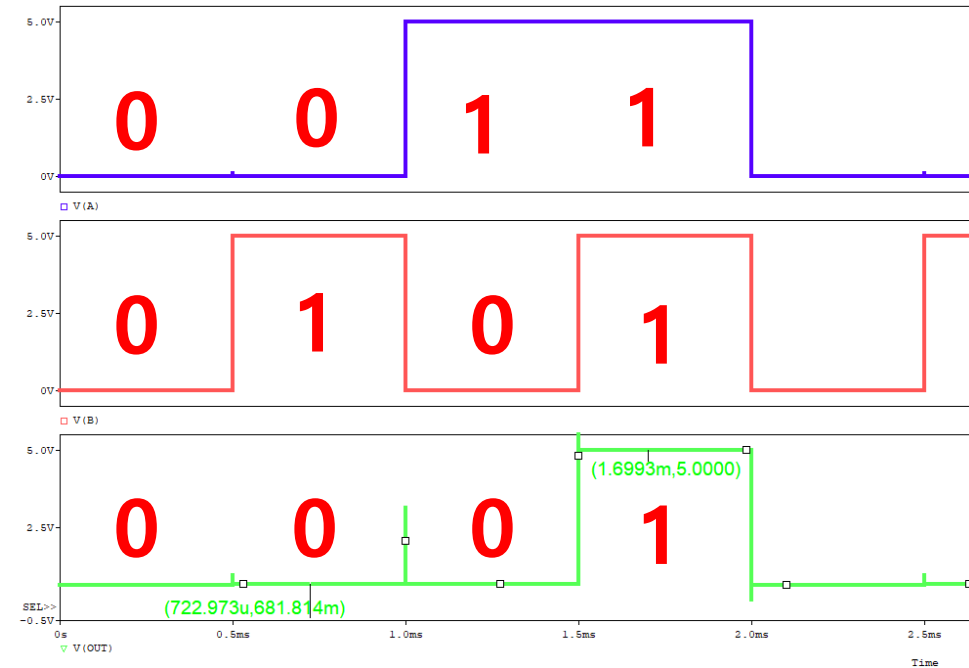
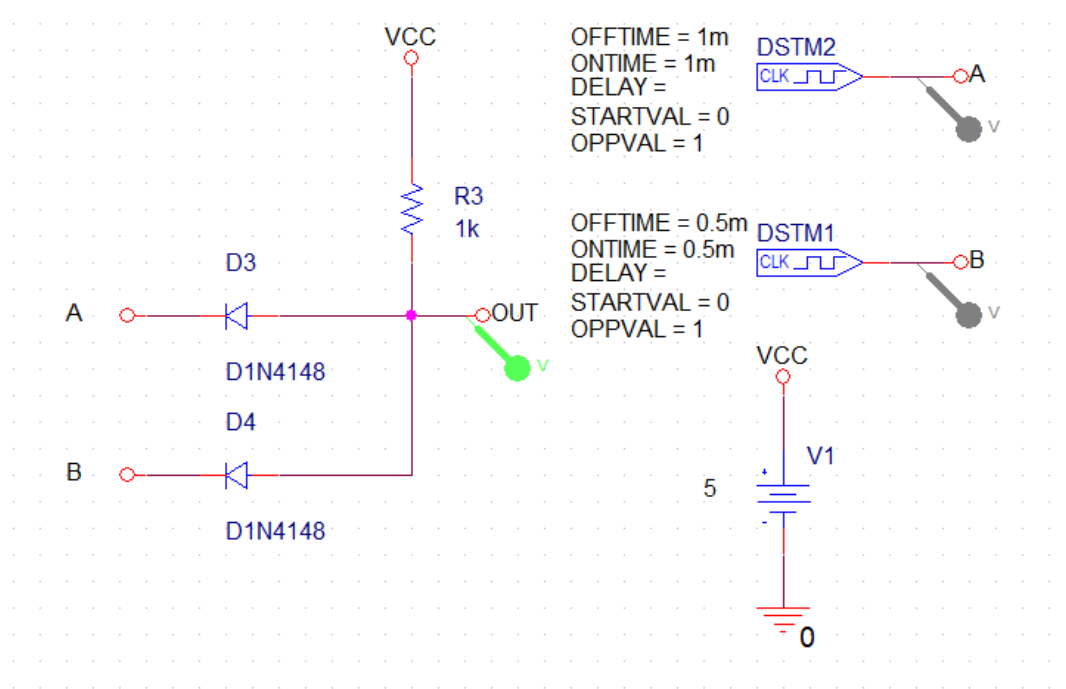


When A or B are High  $\rightarrow$  OUT = H  
If A and B is Low  $\rightarrow$  OUT = L



OR GATE

# 1~(2) AND GATE Design with diodes and Simulation

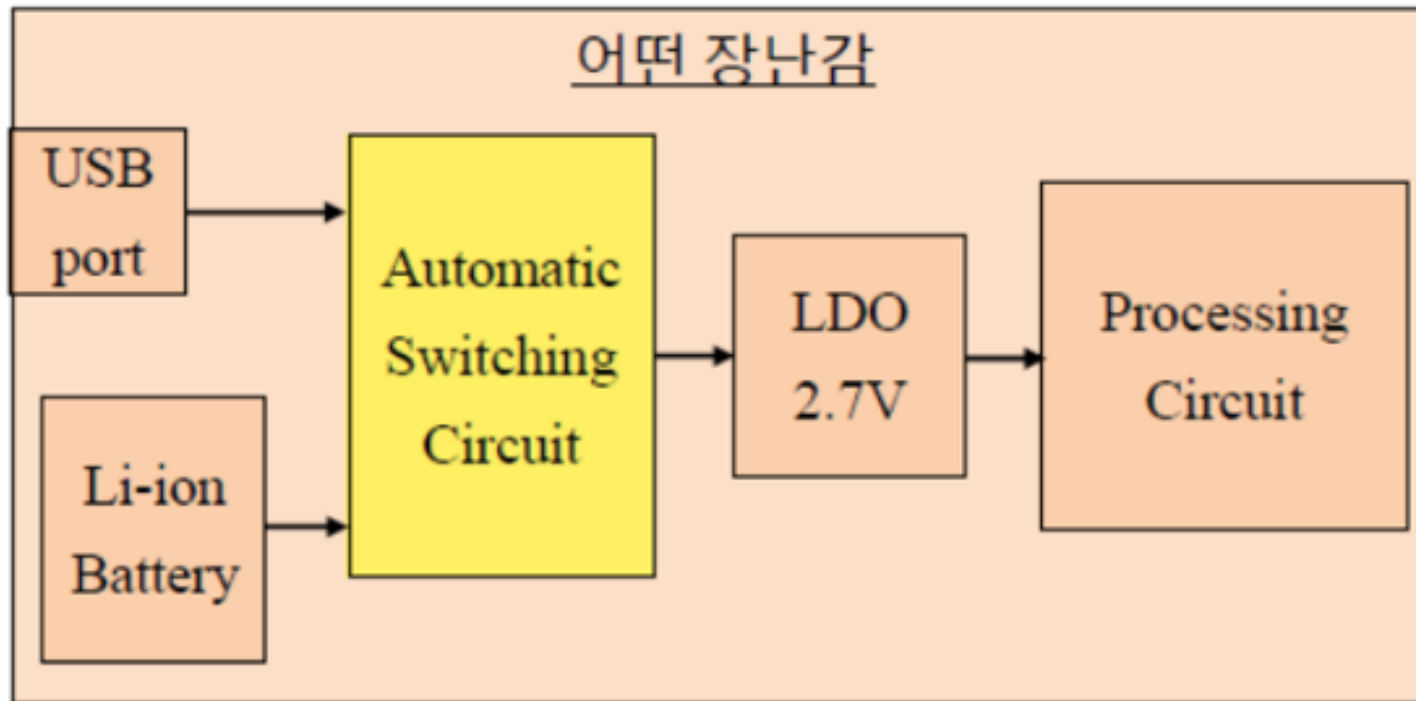


When A and B are High  $\rightarrow$  OUT = H  
If A or B is Low  $\rightarrow$  OUT = L

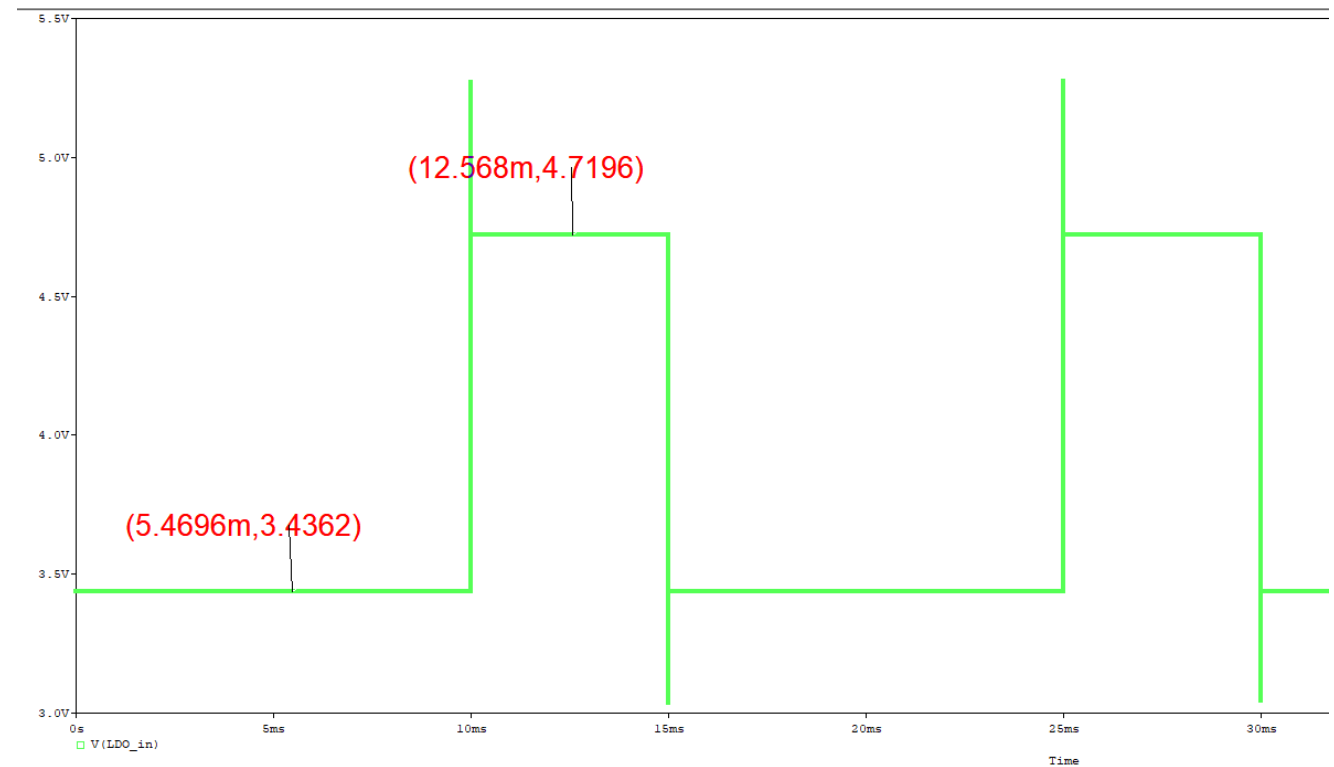
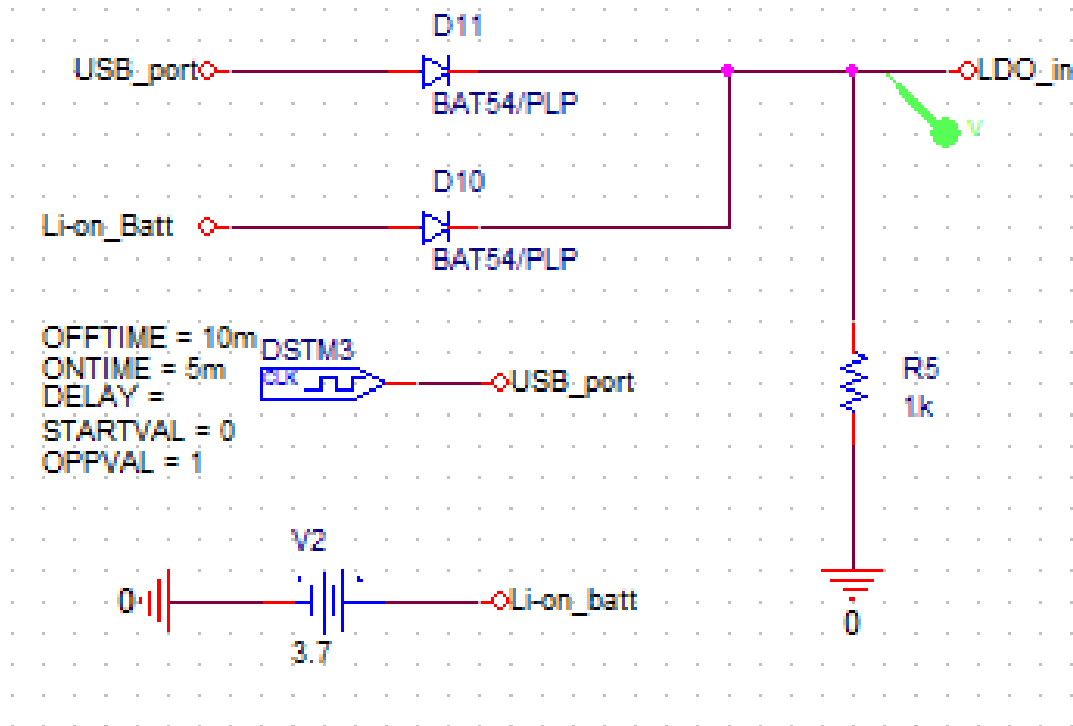


AND GATE

## 2. Automatic Power Switching Circuit Design (input : USB/Li-ion, No Charging)



## 2. Automatic Power Switching Circuit Design

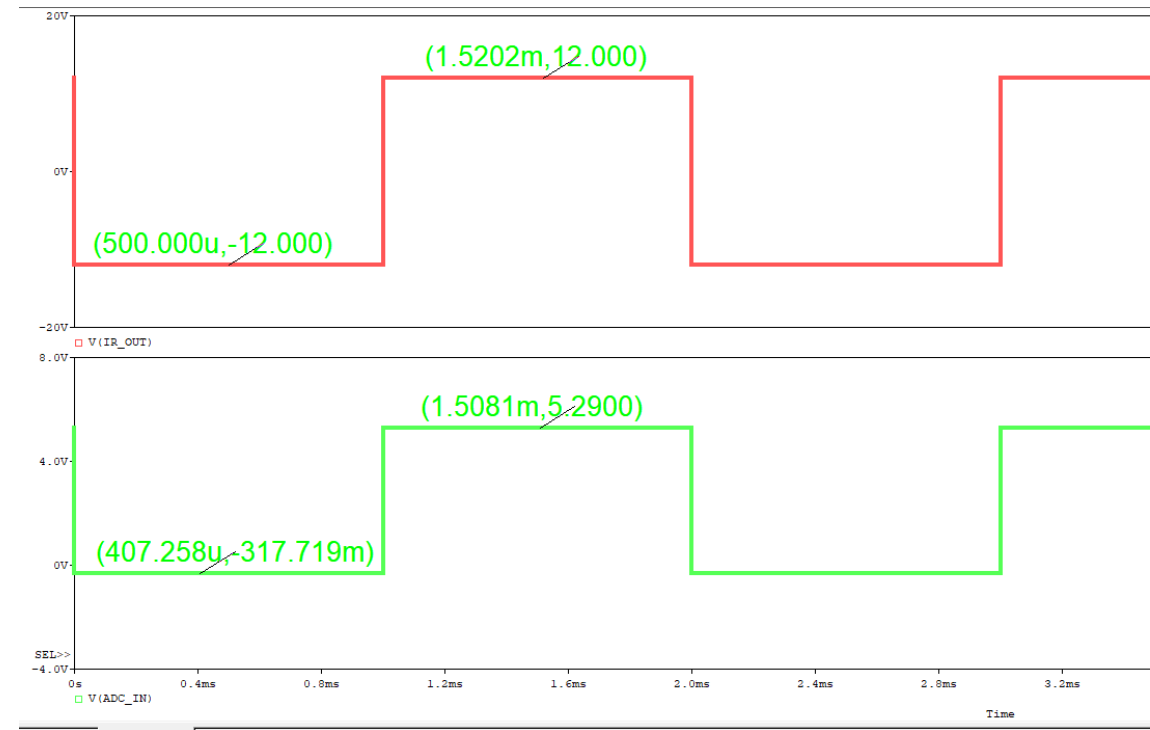
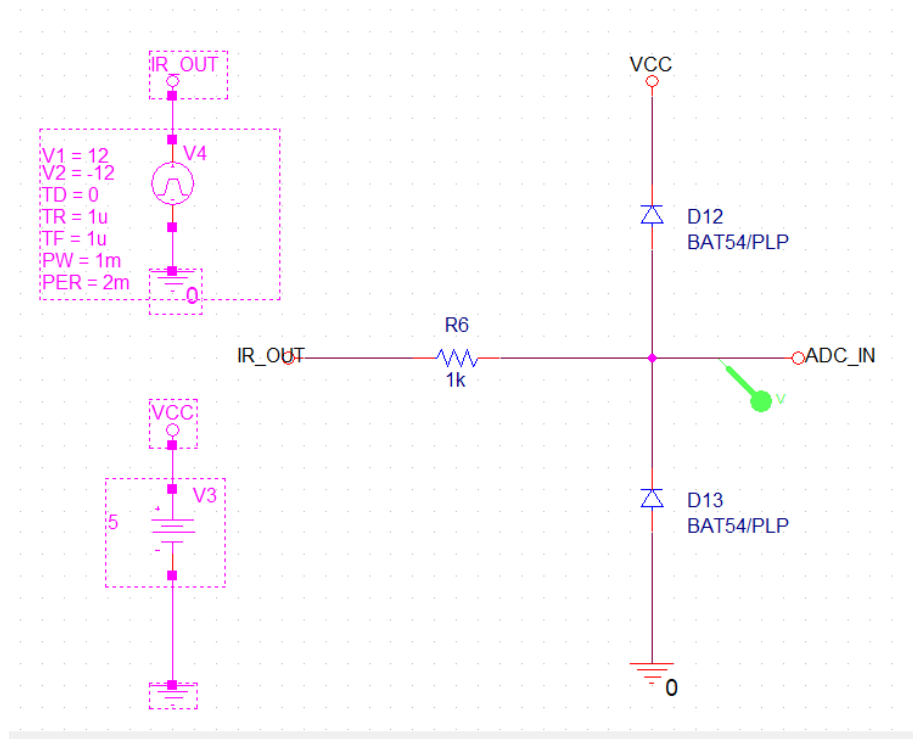
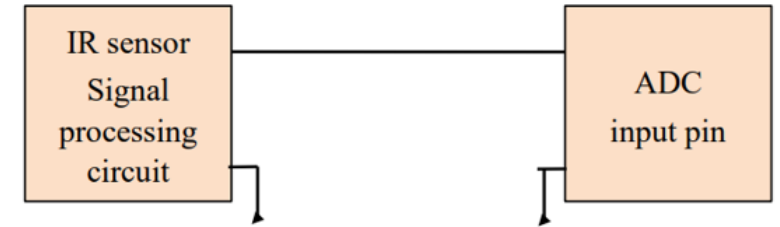


Two-diode power OR-ing

→ VIN (higher source wins): Li-ion (3.0–4.2 V) via BAT54;  
USB (4.6–5.25 V) via 1N4148 (low load) or SS14 (higher load/lower loss).

ESD: place a uni-directional TVS at the USB connector.

### 3. ADC Input Protection Circuit Design (0–5V Range Limit)



-> Use two BAT54 SBDs to rail-clamp the ADC input to VCC and GND, preventing abnormal voltages